# Driver Performance Data

Exploratory Data Analysis Summary

## 1. Project Overview

This report presents the exploratory data analysis of a Metro Manila traffic and road incident dataset. The purpose of the analysis is to prepare the data for a driver-performance monitoring system using PostgreSQL, Laravel, Redis, Next.js, Chart.js, and Metabase.

Because the public dataset does not contain actual individual-driver records, driver IDs, ratings, violations, and behavioral metrics were synthetically generated for educational and demonstration purposes.

## 2. Dataset Overview

The final transformed dataset contains:

| **Information** | **Value** |
|---|---|
| Total records | 17,312 |
| Unique drivers | 100 |
| Date range | August 20, 2018 – December 27, 2020 |
| Average delay | 19.03 minutes |
| Average rating | 4.01 |
| Total accidents | 12,787 |
| Total violations | 34,926 |

The dataset includes the following fields:

- driver_id
- date
- delays_minutes
- behavioral_problems
- violations_count
- accidents_count
- rating
- route_id
- vehicle_id
- shift

## 3. Data Preparation and Cleaning

The following data-quality checks were performed:

| **Cleaning or validation step** | **Action Performed** | **Reason** |
|---|---|---|
| Missing values | Counted missing values in each column | To identify incomplete records |
| Data types | Checked the data type of every column | To ensure dates, numbers, and text were processed correctly |
| Duplicate records | Checked for duplicate rows; none were found | To prevent counting the same incident more than once |
| Coordinates | Replaced invalid (0, 0) latitude and longitude values with NaN | To prevent false locations from being used |
| Time formats | Corrected malformed time values | To correctly derive the driver shift |
| City names | Changed ParaÃ±aque to Parañaque | To prevent the same city from being treated as two categories |
| Whitespace | Removed unnecessary spaces from text values | To combine duplicate categories such as BUS and BUS |
| Direction values | Kept NB, SB, EB, and WB; invalid values were set to NaN | To maintain consistent direction categories |
| Route identifiers | Removed invalid tweet text and rebuilt the route IDs | To ensure route IDs represented locations |
| Numeric values | Checked delays, lanes, accidents, violations, and ratings | To detect impossible or invalid values |

The final transformed CSV was prepared for import into PostgreSQL and Laravel seeding.

## 4. Data Transformation

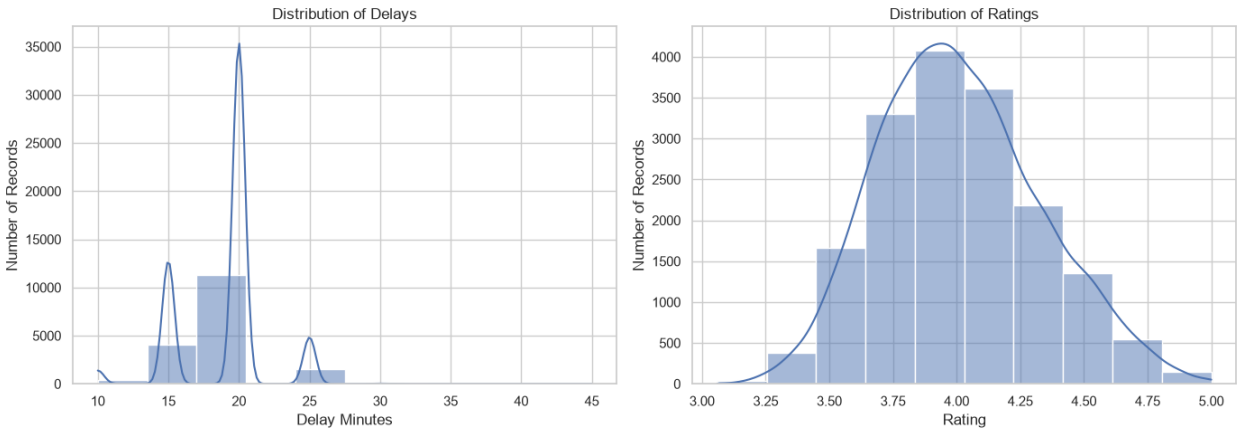
The original traffic incident data was transformed into a driver-performance dataset.

The following fields were derived:

| Field | How it was created | Purpose |
|---|---|---|
| accidents_count | Derived from accident and collision-related incident types | Represents accident frequency |
| delays_minutes | Estimated using incident type and blocked-lane information | Represents delay performance |
| shift | Derived from the recorded time | Groups records by shift |
| route_id | Created from city, location, and direction | Identifies the reported route |
| vehicle_id | Synthetically generated | Provides vehicle identifiers |
| driver_id | Synthetically generated | Supports driver-level analysis |
| behavioral_problems | Synthetically generated for educational purposes | Demonstrates behavioral metrics |
| violations_count | Synthetically generated for educational purposes | Demonstrates violation analysis |
| rating | Synthetically generated for educational purposes | Demonstrates rating analysis |

These derived values are proxies and should not be interpreted as actual driver records.

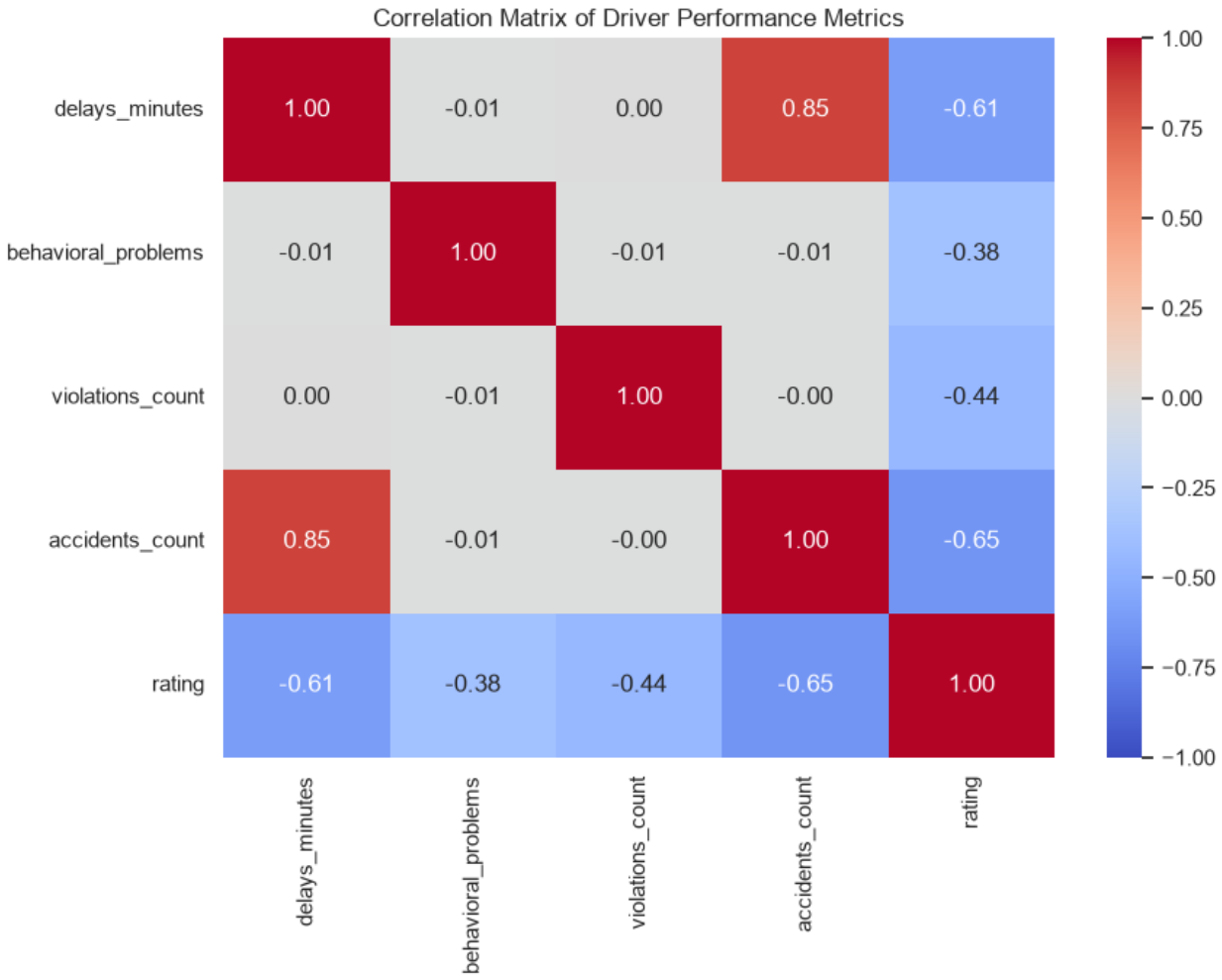
## 5. Distribution Analysis




The average delay was approximately 19.03 minutes. Most delay values were concentrated between 15 and 25 minutes, while higher delay values were less common.

The average driver rating was approximately 4.01 out of 5. Most ratings were within the upper range, while lower ratings appeared less frequently.

## 6. Correlation Analysis




The correlation matrix shows the relationships among delays, behavioral problems, violations, accidents, and ratings.

The strongest positive relationship was between delays_minutes and accidents_count, with a correlation of 0.85. The strongest negative relationship was between accidents_count and rating, with a correlation of -0.65.

Other negative relationships included delays and rating (-0.61), violations and rating (-0.44), and behavioral problems and rating (-0.38).

Because several driver-related variables were synthetically generated, these results demonstrate the analysis workflow but should not be treated as real-world driver-performance conclusions.

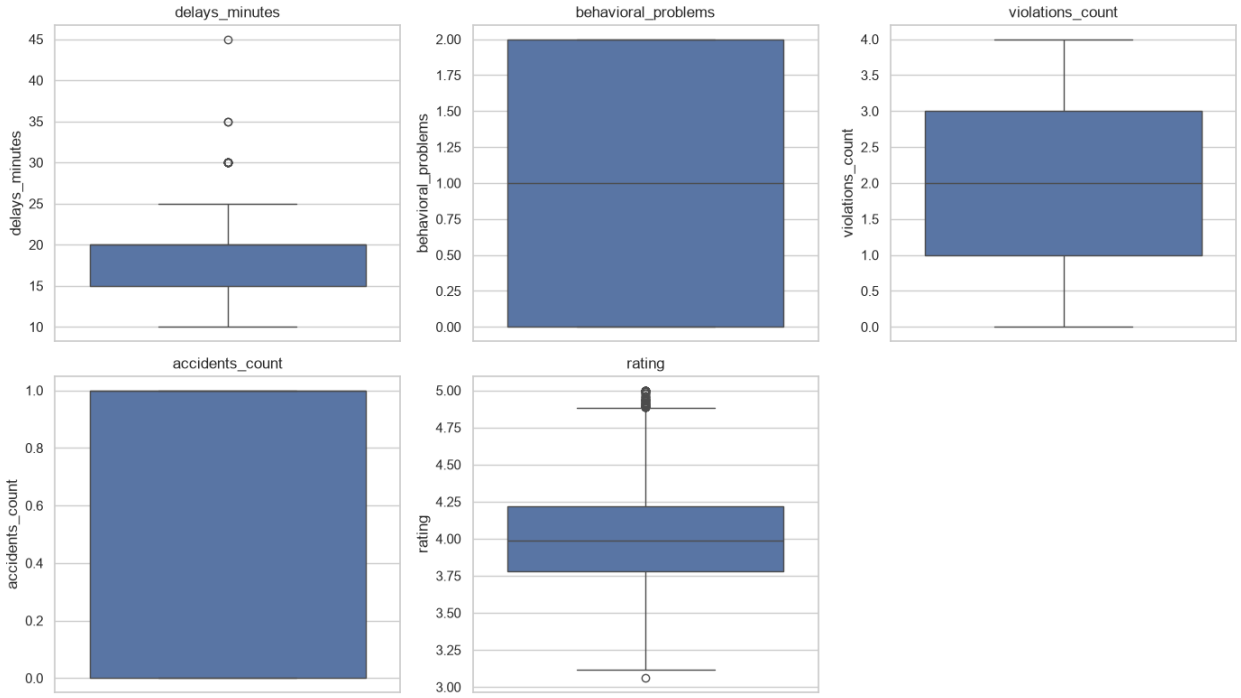
## 7. Weekly Trend Analysis




The highest weekly average delay occurred during the week of December 2, 2018, with approximately 20.05 minutes. Weekly delay values remained relatively close to the overall average, although some weeks showed higher or lower traffic-related delay levels.

The lowest weekly average rating occurred during the week of March 29, 2020, with an average rating of approximately 3.80. The weekly rating trend showed some variation over time, but most weekly averages remained close to 4.00.

## 8. Outlier Analysis




The Interquartile Range method was used to identify unusual values.

| Column | Number of outliers |
|---|---|
| delays_minutes | 13 |
| behavioral_problems | 0 |
| violations_count | 0 |
| accidents_count | 0 |
| rating | 60 |

Outliers were not automatically removed because some unusual values may represent valid incidents or unusual driver-performance records rather than data errors.

The 60 rating outliers were still within the valid rating range of 1.0 to 5.0, so they were retained.

## 9. Key Findings

The main findings from the analysis are:

1. The dataset contains 17,312 records covering more than two years.
2. The average delay was approximately 19.03 minutes.
3. The average rating was approximately 4.01.
4. Weekly delay values remained generally consistent, with the highest weekly average occurring in early December 2018.
5. Weekly ratings were mostly close to 4.00, with the lowest average occurring in late March 2020.
6. A small number of delay values were identified as outliers.
7. Rating outliers were detected but remained within the valid 1.0–5.0 range.
8. The transformed dataset is suitable for database storage, API aggregation, and dashboard visualization.
9. Driver-related metrics are synthetic and should only be used for educational demonstration.

## 10. Recommendations

Based on the analysis, the system should:

- Monitor drivers with consistently high delay values.
- Review drivers with repeated accidents or violations.
- Track rating trends over time.
- Use date and driver filters for more detailed investigations.
- Cache frequently requested metrics using Redis.
- Continue collecting real driver-level data for future analysis.
- Add automated alerts for unusually high accident or violation counts.
- Consider developing a driver-risk score after real data becomes available.

## 11. Limitations

This analysis has several limitations:

- The original public dataset contains traffic incidents, not actual driver records.
- Driver IDs were synthetically generated.
- Ratings, behavioral problems, and violations were synthetically generated.
- Delay values were estimated using traffic incident information.
- The results should not be used for real employee evaluation or disciplinary decisions.
- More reliable conclusions require real driver logs, GPS data, trip records, and verified ratings.

## 12. Conclusion

The exploratory data analysis successfully prepared the traffic incident data for a driver-performance monitoring system.

The final dataset was cleaned, transformed, stored in PostgreSQL, exposed through Laravel API endpoints, cached using Redis, and visualized through Next.js, Chart.js, and Metabase.

The analysis identified delay patterns, rating trends, correlations, and outliers. Although the driver-specific fields are synthetic, the project demonstrates how public traffic data can be transformed into a structured dataset for analytics and dashboard development.